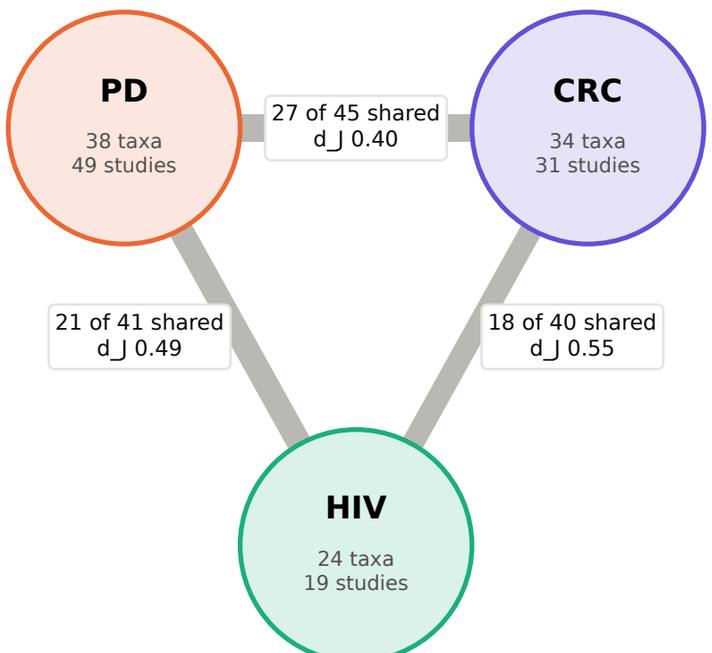
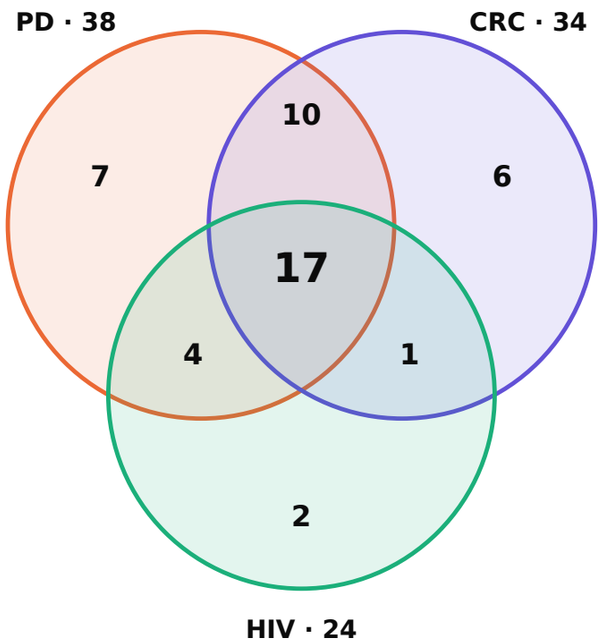
Cellulose-annotated taxa reported decreased in stool overlap across PD, HIV and CRC

Mean pairwise Jaccard distance d_J = 0.479; 17 of 47 taxa appear in all three. Randomly chosen stool studies overlap about as much when their co-reported taxa are kept together (U6, p = 0.19).

A. Shared taxa between each pair

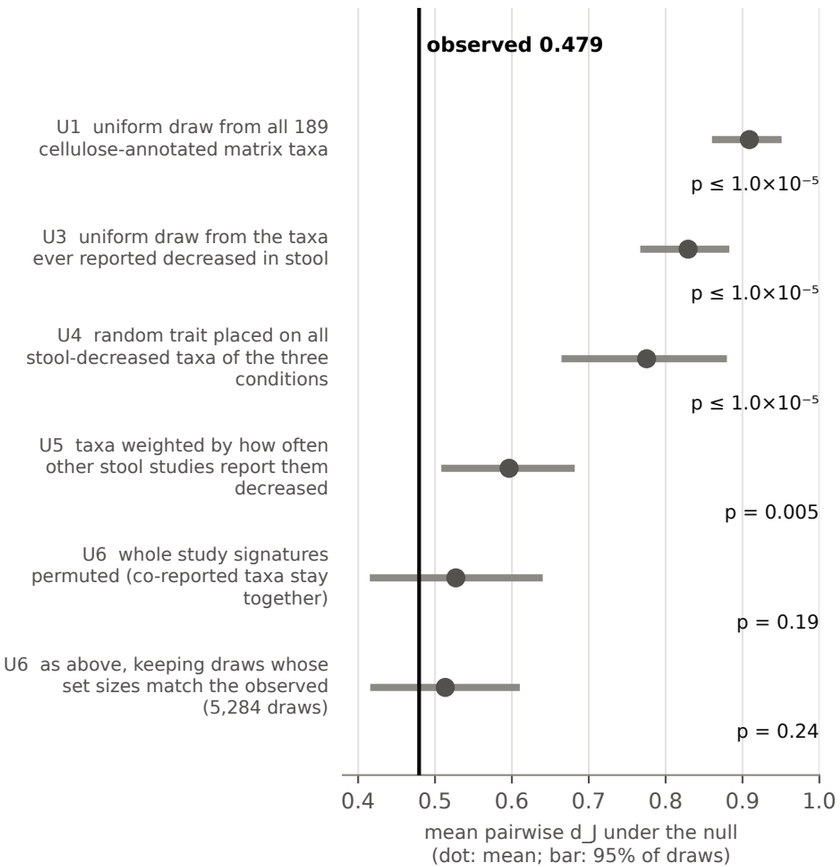


B. Where the 47 taxa fall



Circle areas are not to scale.

C. Is the overlap unusual?

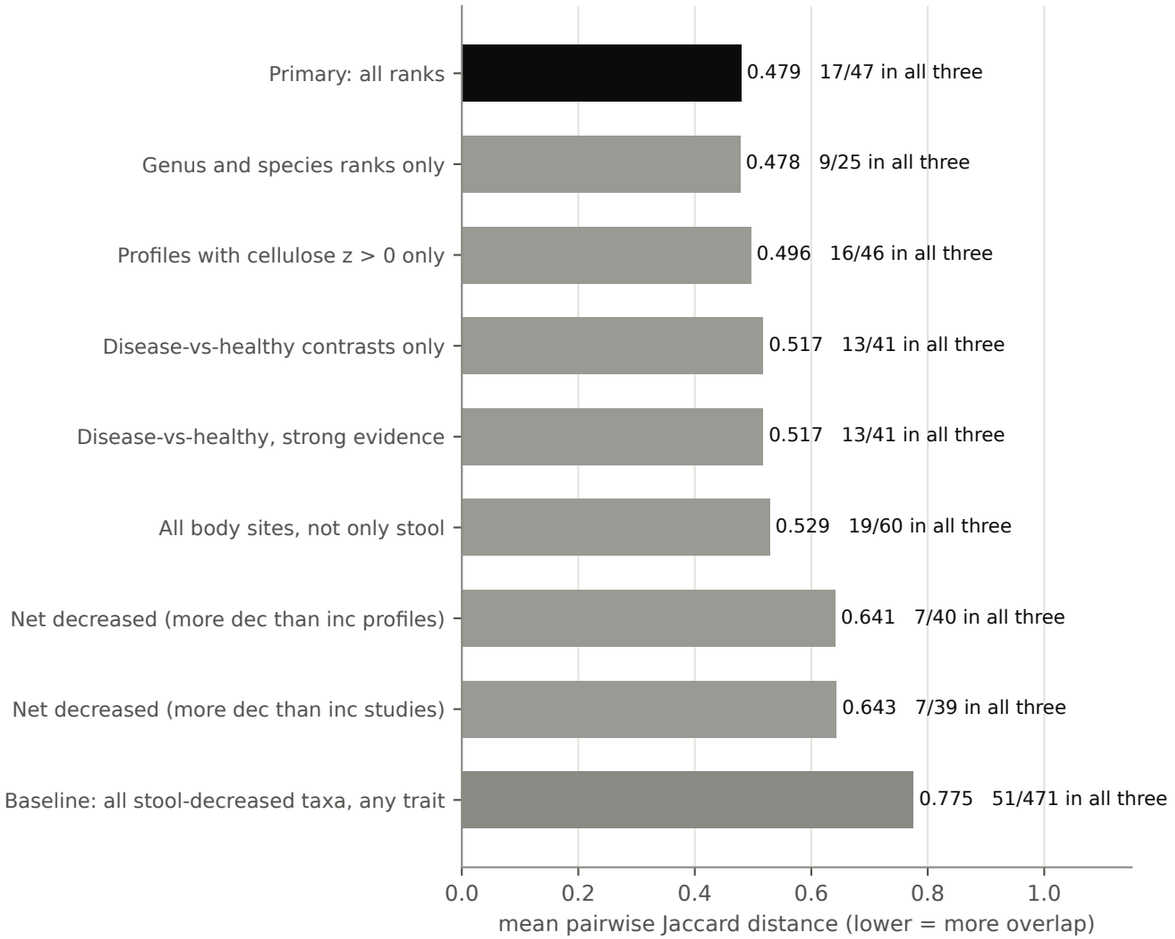


Taxa: KG-Microbe matrix taxa annotated METPO:2000103|GO:0030245 (cellulose catabolic process; any rank, annotations rolled up from descendants) in at least one decreased stool signature of the condition. Overlap counts reported taxonomic labels, not consistent loss: a taxon counts once it is reported decreased in any study. Monte Carlo p from 100,000 draws. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.

The 17 taxa reported decreased in stool in all three conditions: 9 genera and 8 higher-rank nodes

7 of them had more decreased than increased stool studies in every condition. Keeping only genus and species ranks gives mean d_J = 0.478, close to the primary 0.479.

A. Mean pairwise d_J under each set definition



B. Stool studies reporting each core taxon ↓ or ↑

		PD	HIV	CRC	net ↓ in all 3
Roseburia	genus	20↓ 4↑	2↓ 0↑	6↓ 0↑	✓
Blautia	genus	13↓ 2↑	1↓ 3↑	6↓ 1↑	
Clostridium	genus	7↓ 3↑	5↓ 2↑	6↓ 3↑	✓
Bacteroides	genus	6↓ 3↑	8↓ 0↑	2↓ 2↑	
Lachnospira	genus	5↓ 2↑	3↓ 0↑	6↓ 1↑	✓
Ruminococcus	genus	6↓ 2↑	4↓ 0↑	4↓ 3↑	✓
Anaerostipes	genus	2↓ 1↑	3↓ 0↑	3↓ 1↑	✓
Eubacterium	genus	1↓ 1↑	1↓ 1↑	3↓ 2↑	
Butyrivibrio	genus	2↓ 2↑	1↓ 2↑	1↓ 0↑	
Lachnospiraceae	family	20↓ 7↑	5↓ 3↑	9↓ 5↑	✓
Bacillota	phylum	8↓ 5↑	4↓ 2↑	4↓ 1↑	✓
Clostridia	class	5↓ 7↑	4↓ 0↑	7↓ 1↑	
Oscillospiraceae	family	8↓ 13↑	4↓ 0↑	2↓ 1↑	
Clostridiaceae	family	8↓ 2↑	3↓ 0↑	2↓ 2↑	
Eubacteriales	order	5↓ 6↑	4↓ 1↑	4↓ 1↑	
Bacteroidia	class	5↓ 2↑	2↓ 0↑	1↓ 4↑	
Bacteroidota	phylum	3↓ 0↑	3↓ 1↑	1↓ 4↑	

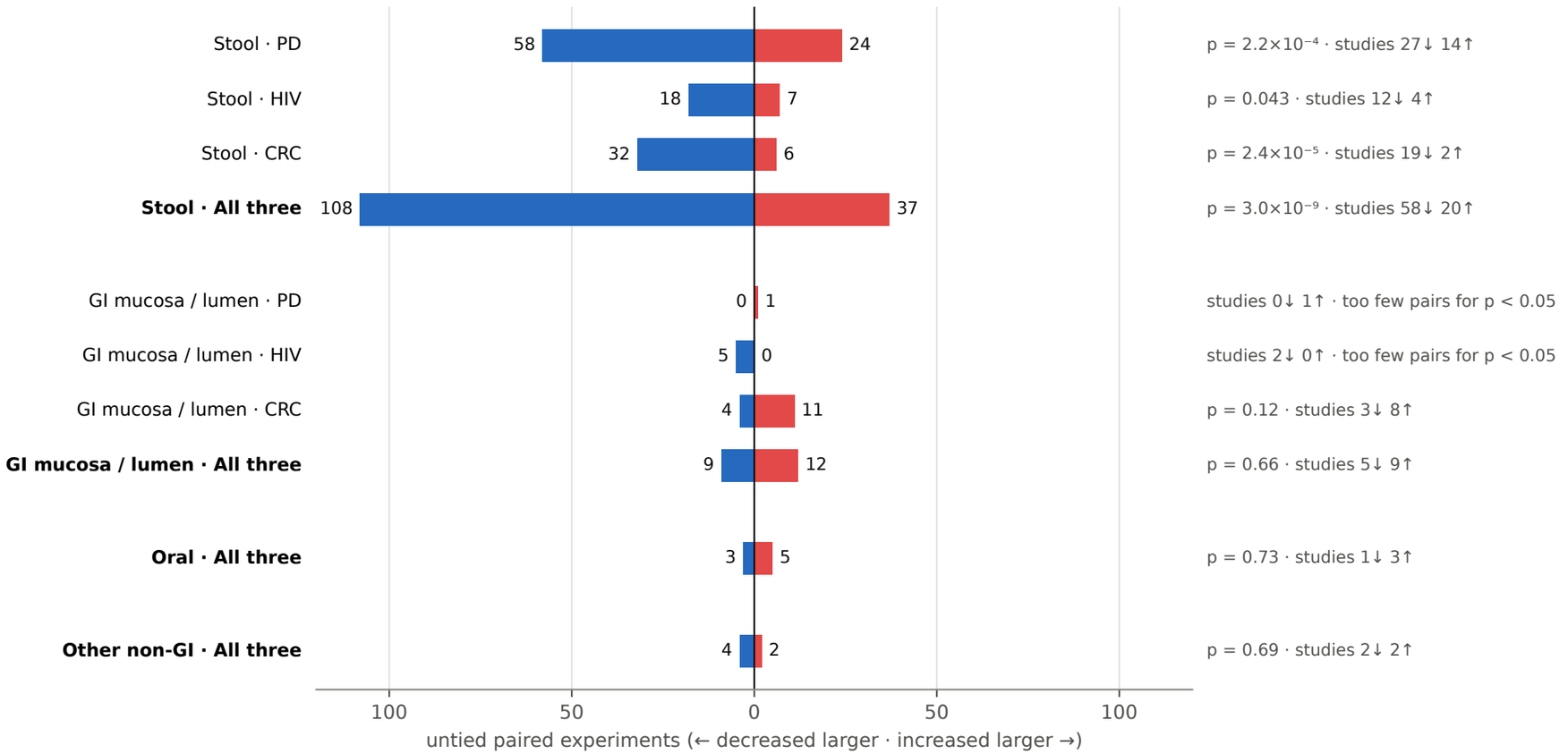
Dot: blue when ↓ studies outnumber ↑, red when ↑ outnumber ↓, grey when equal. Upper block: genera; lower block: family, order, class and phylum nodes, whose annotations are rolled up from descendant taxa. Net ↓ = more decreased than increased studies in every condition.

A taxon is in a condition's set when at least one decreased stool signature reports it; the counts in B show that some core taxa are also reported increased, sometimes more often (for example Oscillospiraceae in PD and Bacteroidota in CRC). PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.

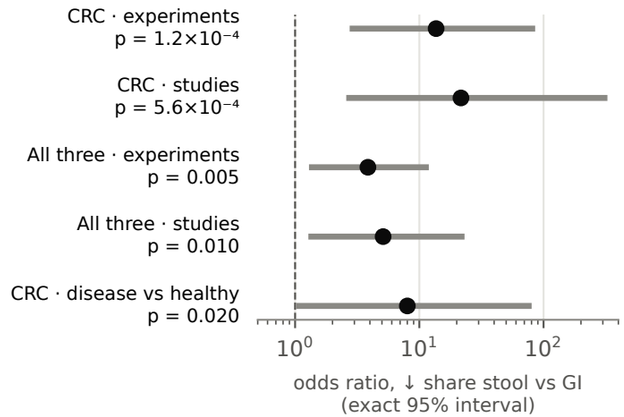
The decreased-over-increased shift is clearest in stool; other sites have too few comparisons to judge

Across PD, HIV and CRC stool, 108 of 145 untied paired experiments had the larger cellulose annotation share in the decreased signature ($p = 3.0\times10^{-9}$); outside stool most strata have fewer than six untied pairs.

A. Paired experiments: decreased signature larger (↓) or smaller (↑)



B. Stool vs GI mucosa / lumen



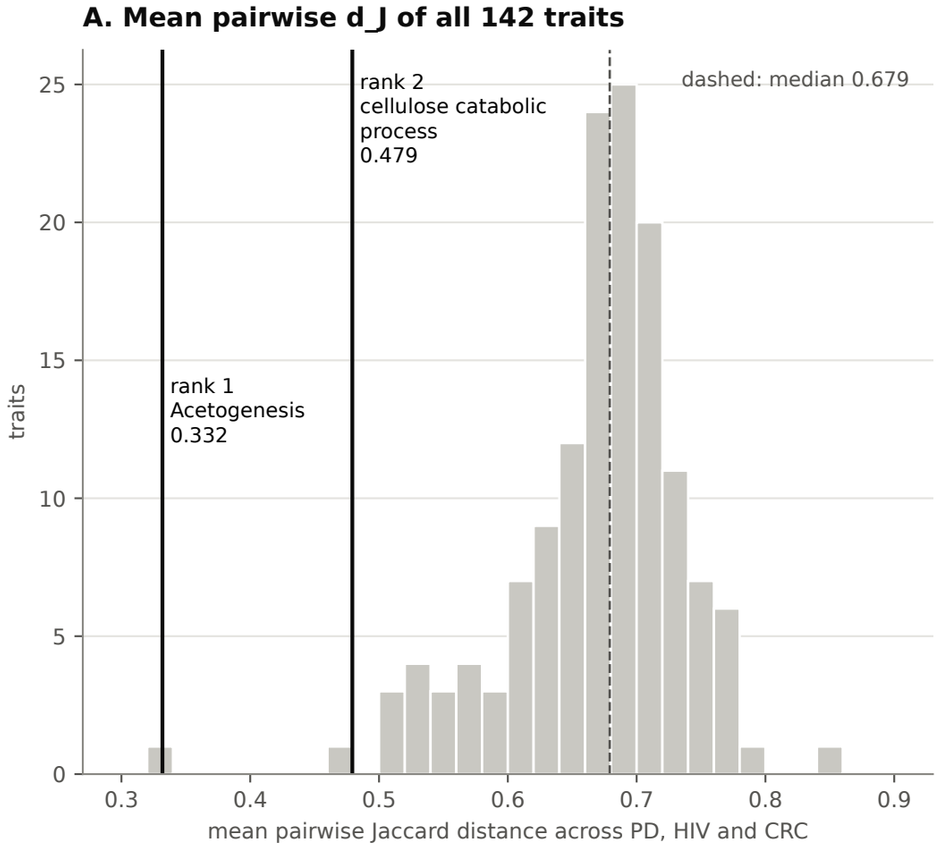
C. Decreased-signature studies by site

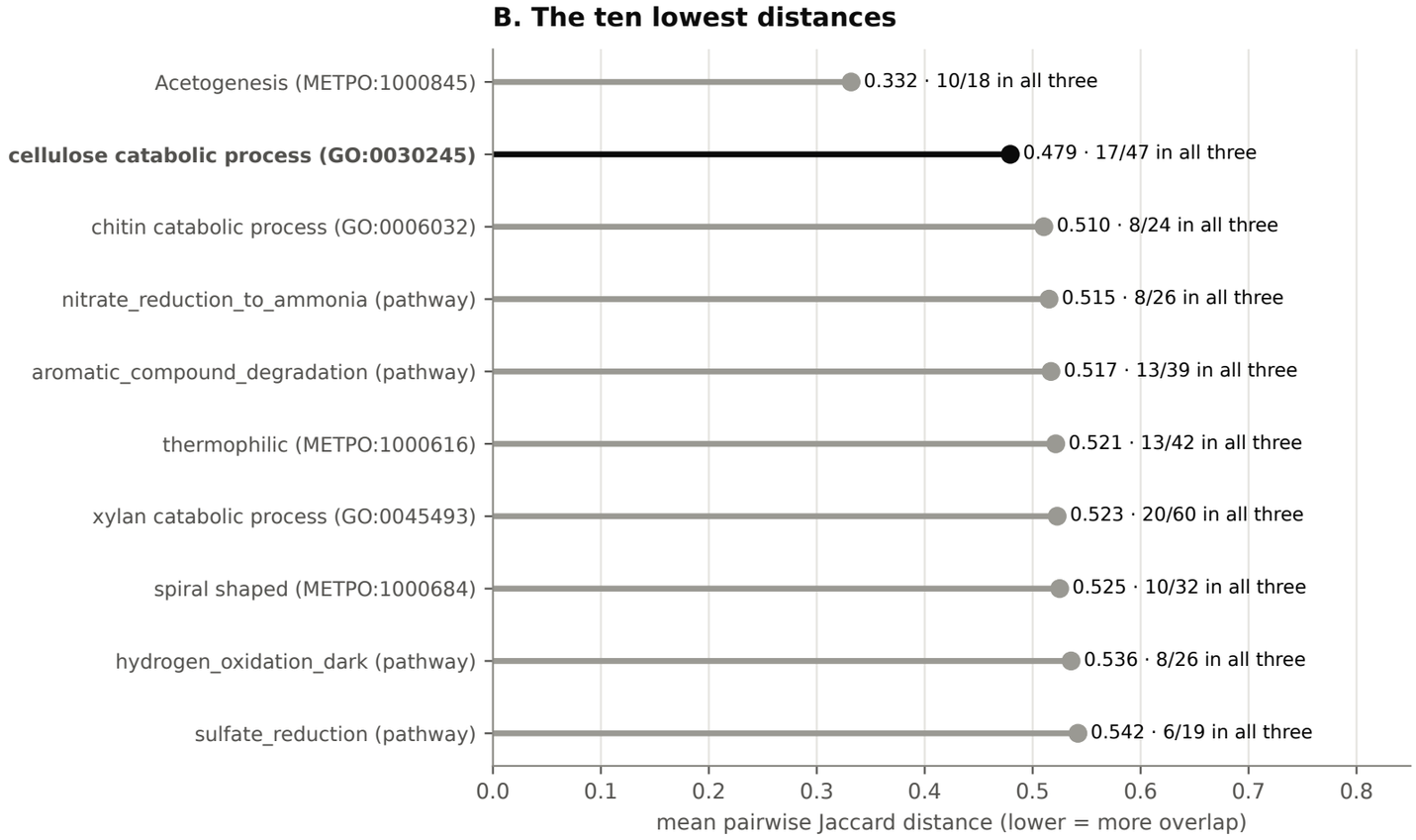
	PD	HIV	CRC
Stool	49	19	31
GI mucosa / lumen	2	4	14
Oral	2	3	4
Other non-GI	3	3	0

Paired experiment: one BugSigDB experiment with both an increased and a decreased signature; the share compared is cellulose annotations over all retained non-MediaDive annotations. Ties excluded. Study rows take the sign of each study's mean paired difference. The Fisher tests treat units as independent; a few studies sampled both compartments, and the sites differ in sampling effort (C), so a difference between compartments is not evidence of anatomical specificity. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.

Among 142 non-MediaDive traits, cellulose catabolic process has the second-lowest cross-condition Jaccard distance

Traits with at least 10 stool-decreased taxa in each condition. Rank 2 of 142, after Acetogenesis (METPO:1000845). Ranks describe this corpus; they are not a test, and traits share taxa.





Trait names are the object_name labels in data/derived/taxa_trait_matrix_columns.tsv. Excluded: MediaDive growth-medium columns (METPO:2000517|mediadive.medium:*). Several listed traits are morphology or temperature classes, not metabolic processes. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.